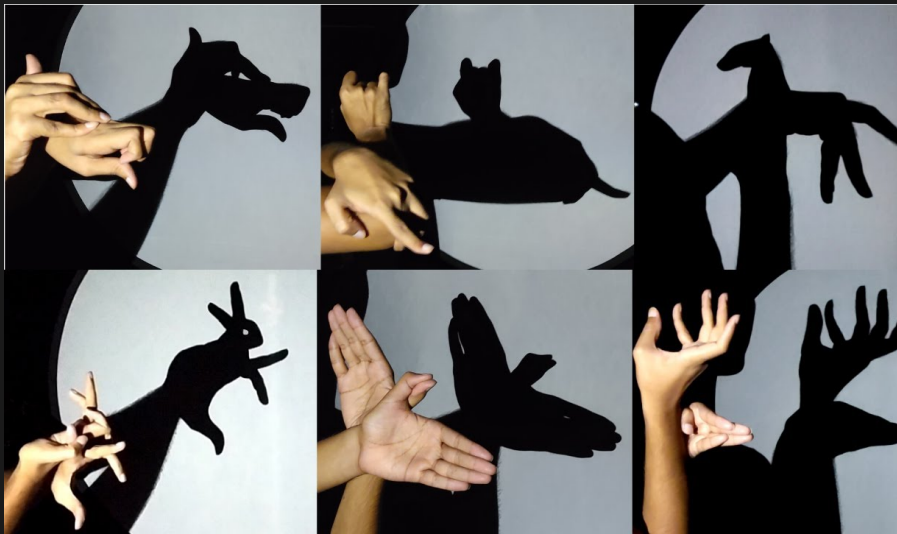


Latent Structures and PCA

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Lecture 04.1 (v4.0.0)



<https://www.youtube.com/watch?v=t-oTvYGLYPw>

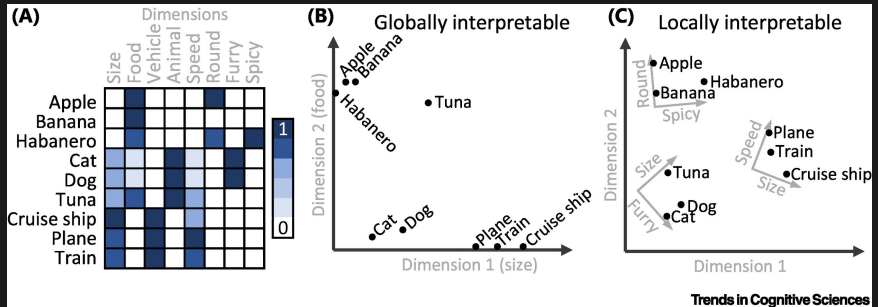
Questions

- ▶ How can we understand high dimensional objects with low-dimensional visual tools?
- ▶ Is an Eigenvector space a latent space?
- ▶ Are parameters of a model a latent space?
- ▶ How are these to be used in Data Science?

Latent Structures

- ▶ It is often useful to think of data X being represented in some (usually lower-dimensional) space θ .
- ▶ θ is the **latent space** for X .
- ▶ Examples:
 - ▶ Parameters: $X_i \sim f(\theta_i)$ for some model f
 - ▶ Kernel representation: $X_i = \sum_{j=1}^K K(\theta_i)$
 - ▶ Factor analysis
 - ▶ Spectral decomposition, Principal Components Analysis, Singular Value Decomposition

Structure of Latent Structures



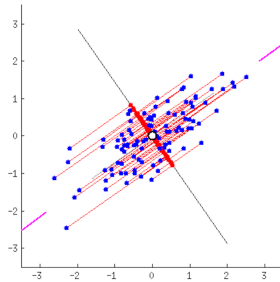
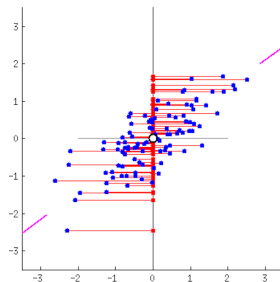
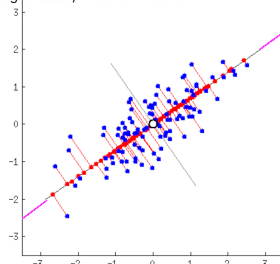
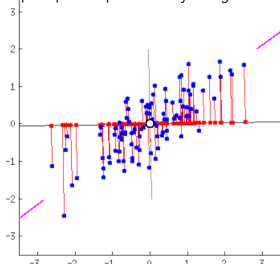
The Dimensions of dimensionality

Latent Structure

- ▶ What makes a **latent space** instead of a parameterisation is the modelling done on that space.
- ▶ i.e. it is constructed to *mean something*
- ▶ If it is done with the intention of making *similar data be close* then we might call this an embedding
- ▶ Much care is needed around the words “similar” and “close”!

PCA Example from Stack Exchange

<https://stats.stackexchange.com/questions/2691/making-sense-of-principal-component-analysis-eigenvectors-eigenvalues/140579#140579>



Principal Components Analysis

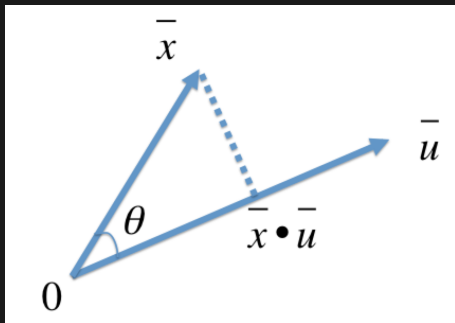
- ▶ PCA is the baseline factor analyses
- ▶ Let $X \in \mathcal{R}^{n \times k}$. How do we find “the main directions” in which X varies?
- ▶ If $C = XX^T$ is the covariance matrix, we will look for solutions of the form:

$$CU = \Lambda U$$

- ▶ Where Λ is a diagonal matrix of **eigenvalues** sorted by magnitude, and
 - ▶ **eigenvectors** $U \in \mathcal{R}^{n \times d}$ form an orthonormal basis.
- ▶ ... how would we compute it?
- ▶ ... and why is this a good idea?

One dimensional projection

- ▶ We will project $\vec{x}$ onto a subspace U .



- ▶ In one dimension this is a line defined by $\vec{u}$ of unit length through the origin.
- ▶ The projection of $\vec{x}_i$ onto $\vec{u}$ is $(\vec{x}_i \cdot \vec{u})$ in the new coordinate system.
- ▶ Recall $\vec{x} \cdot \vec{y} = |\vec{x}||\vec{y}| \cos(\theta)$

One dimensional projection

- Averaging over all vectors:

$$MSE(\vec{u}) = \frac{1}{n} \sum_{i=1}^n \vec{x}_i \cdot \vec{x}_i - (\vec{u} \cdot \vec{x}_i)^2,$$

- But the first term is constant, so we therefore are seeking to **maximise**:

$$\frac{1}{n} \sum_{i=1}^n (\vec{u} \cdot \vec{x}_i)^2,$$

- This is the second moment of $\vec{u} \cdot \vec{x}_i$ which can be written as:

$$\mathbb{E}(\vec{u} \cdot \vec{x}_i)^2 + \text{Var}(\vec{u} \cdot \vec{x}_i).$$

- However, the mean of the projection is zero by construction so **minimising the MSE is equivalent to maximising the variance explained.**

Multiple dimensions

- ▶ To work with multiple dimensions:
 - ▶ replace the single vector projection $(\vec{u} \cdot \vec{x}_i)\vec{u}$ with a sum over all dimensions $\sum_{k=1}^K (\vec{u}_k \cdot \vec{x}_i)\vec{u}_k$.
- ▶ Or in matrix notation, $\vec{U} = XU$.
- ▶ It is straightforward to show that the cross terms cancel out,
 - ▶ due to the orthogonality constraint $\vec{u}_k \cdot \vec{u}_j = 0$ for $k \neq j$.

Maximising variance

- ▶ Using matrix notation,

$$\text{Var}(\vec{U}) = \frac{1}{n-1} (\mathbf{X}\mathbf{U})^T (\mathbf{X}\mathbf{U}) \quad (1)$$

$$= \mathbf{U}^T \frac{\mathbf{X}^T \mathbf{X}}{n-1} \mathbf{U} \quad (2)$$

$$= \mathbf{U}^T \mathbf{C} \mathbf{U} \quad (3)$$

- ▶ Where $\mathbf{C}$ is the covariance.
- ▶ In the discussion, we constrain the search over $\mathbf{U}$ to look for unit vectors with a **Lagrange multiplier** λ , which allows unconstrained optimisation of:

$$\mathbf{u}^T \mathbf{C} \mathbf{u} = \lambda \Rightarrow \mathbf{C} \mathbf{u} = \lambda \mathbf{u}$$

Interpretation

- ▶ Our basis $\mathbf{U}$ is **orthonormal**, and
- ▶ As $\mathbf{C}$ is a covariance matrix, it is symmetric.
- ▶ Therefore the **eigenvectors are orthogonal**.
- ▶ The **eigenvalue** (i.e. λ_k) is the **variance explained** by the k -th PC.
- ▶ The proportion of **variance explained** by K PCs is
$$R^2 = \frac{\sum_{k=1}^K \lambda_k}{\sum_{m=1}^M \lambda_m}.$$
- ▶ If the data are really in a K dimensional linear subspace, the eigenvalues beyond that should be 0.
- ▶ The mean usually shows up in PC1 if you leave it in.

Singular value decomposition

- ▶ PCA and SVD are related:

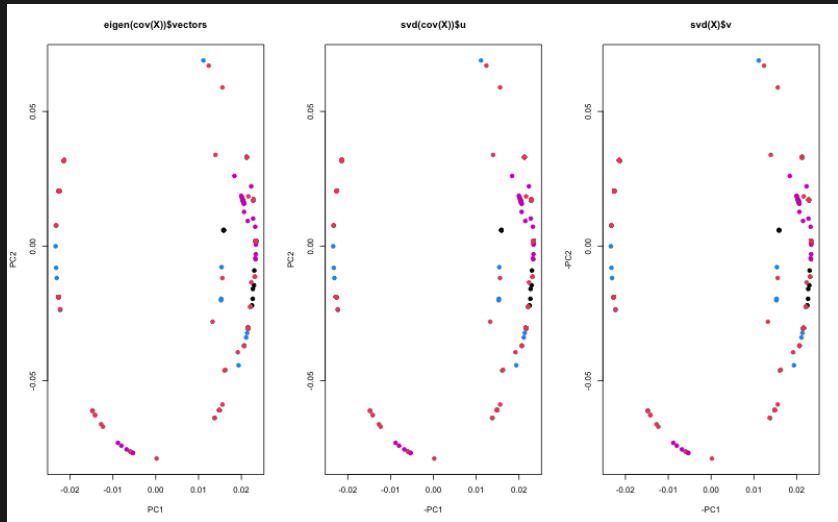
$$\text{SVD}(X) = UDV^T$$

- ▶ and therefore

$$\frac{1}{n-1}X^TX = \frac{1}{n-1}VD^TU^TUDV^T = V\frac{D^TD}{n-1}V^T = V\Lambda V^T$$

- ▶ where $\Lambda = D^TD/(n-1)$ are both diagonal matrices.

PCA/SVD/Covariance plots

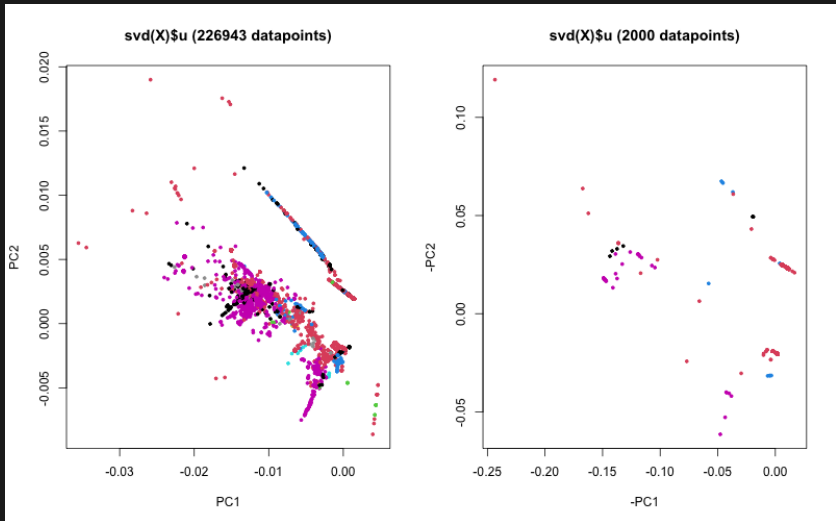


(see provided Rscripts)

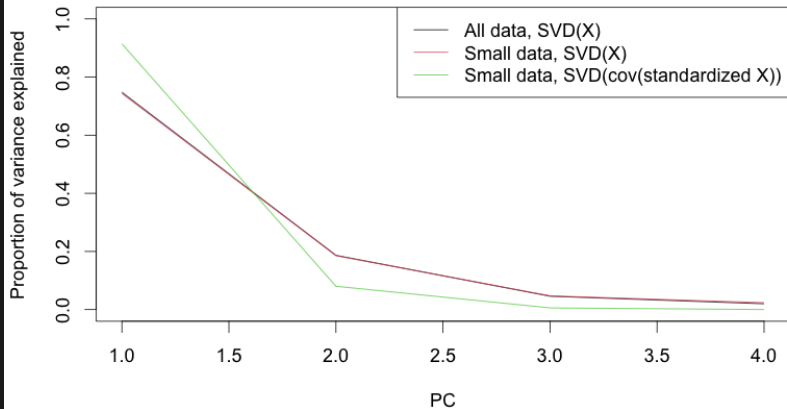
Some notes

- ▶ Here, mean centring X to $\tilde{X}$ is weird; its re-weighting features differently for each observation!
- ▶ $SVD(X)$ and $SVD(\text{Cov}(\tilde{X}))$ contain the same structure, elucidated differently
- ▶ When there are many fewer features than observations, working with X directly is **much faster**
- ▶ If the number of features is higher than the number of datapoints, working with Covariance makes sense.

Sampling data



Variance Explained



Reflection

- ▶ You should:
 - ▶ Be able to intuitively explain PCA, and perform simple calculations using it
 - ▶ Be able to relate PCA to SVD both mathematically and intuitively
 - ▶ Be able to deploy either appropriately on real data
- ▶ What is Spectral Decomposition doing? Why is it a good idea?
- ▶ How is it different to a model?
- ▶ What might it mean that two datapoints are “close” in a PCA plot?
- ▶ What might go wrong when making a 2-D PCA plot?

Signposting

- ▶ We will look at **clustering** - both on the raw data but also on PCs, which can often be used to avoid discrete features posing a problem as well as being efficient.
- ▶ We'll explore PCA in the Workshop.
- ▶ Further reading could include:
 - ▶ **Cosma Shalizi's Advanced Data Analysis, Lecture 18**
 - ▶ **Boyd and Vandenberghe: Convex Optimization** is an excellent and thorough resource.
 - ▶ I showed **Kalman: Leveling with Lagrange: An Alternate View of Constrained Optimization**